

Yikai Tang

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Education

Shanghai Jiao Tong University (SJTU)

Sep 2022 – Present

Advisor: Prof. Yunbo Wang

B.E. in Computer Science and Technology (IEEE Honor Class)

- GPA: 3.8
- 27 A/A+ Courses: Design and Analysis of Algorithms, Operating System, Discrete Mathematics, etc.
- GRE: V160, Q168; TOEFL: 110

University of California, Berkeley

Jan 2025 – May 2025

Advisor: Prof. Jitendra Malik

Visiting Student, BGA Program

- CS 184: Foundations of Computer Graphics — A
- CS 61A: Structure and Interpretation of Computer Programs — A
- EECS 126: Probability and Random Processes — A-

Publications

DIPOLE

BAIR, UC Berkeley | Mar 2025 – Sep 2025

Fusing Vision and Geometry for Robust Visuomotor Generalization

Y. Tang*, H. Geng*, J. Jia, Y. Hu, S. Zang, J. Yang, P. Abbeel, J. Malik

- Proposed a **modality-aware dropout mechanism** enabling robust RGB–point-cloud fusion within diffusion policies.
- Built a unified visual–geometric latent space that improves generalization across lighting, materials, viewpoints, and object configurations.
- Demonstrated strong, robust, state-of-the-art performance through geometry-informed policy learning.

ISF-VLA

IROS 2026

textit | BAIR, UC Berkeley | Sep 2025 – Nov 2025

An Implicit 3D Vision-Language-Action Model Capitalizing on 3D-Aware Vision Foundation Models through Implicit Spatial Fusion

S. Zang*, S. Wei*, H. Geng, Y. Tang, B. An, S. Jayavelu, P. Abbeel, J. Malik

- Introduced **implicit 3D spatial fusion** to integrate geometric cues into RGB–point-cloud fusion for VLA.
- Achieved significant gains in cross-scene and cross-viewpoint robustness through 3D-grounded multi-modal representation learning.
- Enabled stable and expressive fusion for large pretrained visual–language backbones in manipulation.

RoST3R

BAIR, UC Berkeley | Mar 2025 – Sep 2025

Robot-Aware Dynamic 3D Reconstruction from RGB Input for Robotic Manipulation

Y. Han, H. Geng, J. Zhang, Z. Chen, Y. Tang, C. Cheng, P. Li, T. Darrell, P. Abbeel, J. Malik, Y. Wang

- Developed a dynamic, metric-scale 3D reconstruction pipeline from monocular RGB tailored for robotic manipulation.
- Enabled consistent 3D scene understanding with temporally stable geometry beneficial for downstream control.
- Delivered robot-aware reconstruction that integrates motion cues for improved dynamic-scene fidelity.

Research Experience

BAIR, UC Berkeley

Mar 2025 – Nov 2025

Advisor: Prof. Jitendra Malik

DIPOLE* — Multimodal diffusion policy with modality-aware fusion

Mar 2025 – Sep 2025

RoST3R — Robot-aware dynamic 3D reconstruction from RGB

Mar 2025 – Sep 2025

ISF-VLA — Implicit 3D spatial fusion for VLA models

Sep 2025 – Nov 2025

Yunbo Lab, SJTU

Oct 2023 – Dec 2024

Advisor: Prof. Yunbo Wang

DeformSDF* — Physics-aware SDF reconstruction for highly deformable objects

Aug 2024 – Dec 2024

Intuitive Physics Fluids — Multi-view real-world fluid reconstruction

Oct 2023 – Jul 2024

*First / Co-First Author

Projects

Game Design with olcPixelGameEngine

Jun 2023 – Jul 2023

Advisor: Yuting Wang

Tech: C++

- Designed and implemented an SLG game with custom UI and unit systems.
- Implemented pixel-level visual effects for movement, attacks, and explosions.

News Crawler and Search Engine

Dec 2023 – Jan 2024

Advisor: Ya Zhang

Tech: Flask, BeautifulSoup, Lucene, Face_Recognition

- Built a multithreaded crawler collecting 10k+ news articles and images.
- Implemented a Lucene-based search backend with a Flask web interface.

Structured Macro Support for WhileDB

Dec 2024 – Jan 2025

Advisor: Qinxiang Cao

Tech: Flex, Bison, C#

- Extended WhileDB with structured macros and function calls while preserving AST integrity.

Honors and Awards

SJTU Merit Scholarship for Academic Excellence (Category C) — 2023

Class Student Leader in Academics — 2022

Professional Service

Workshop Organizer: [NeurIPS 2026] Embodied World Models for Decision Making [↗](#)

Leadership and Activities

Debate Team, SJTU — Competed in 30+ tournaments; mentored 20+ members.

Technical Skills

Python, C++, Coq; PyTorch, ROS2, Transformers, Open3D; Linux, Git